

Regression Prediction of all Customers' Next Car-service Date and Odometer

Project Objective

To predict all customers' next car-service visit date and odometer reading, with the goal of limiting the deviation in predicted dates to within one month, thereby enabling the business dept to formulate data-driven marketing strategies.

Challenges

- 1. Low-frequency and low-loyalty user behavior, making patterns difficult to capture.
- 2. High volume of missing and noisy data, which significantly increases the complexity of achieving a prediction accuracy within a one-month deviation.
- 3. Designing an effective predictive model while preventing data leakage, ensuring both reliability and robustness.

Model Results

模型 LGB Model	规则全量 Statistics Rule Model
1 tongjil(dt, '天数偏差分类1')	1 tongjil(rule, '天数偏差分类1')
天数偏差分类1 fenzi fenmu ratio	天数偏差分类1 fenzi fenmu ratio
0 30-60天 68418 316906 0.215894	0 30-60天 99440 480498 0.206952
1 30天以内 126972 316906 0.400661	1 30天以内 146486 480498 0.304863
2 60-90天 40412 316906 0.127520	2 60-90天 63439 480498 0.132028
3 90天以上 81104 316906 0.255924	3 90天以上 171133 480498 0.356158

模型 LGB Model	规则全量 Statistics Rule Model
1 tongji2(dt, '公里数偏差分类1')	1 tongji2(rule, '公里数偏差分类1')
公里数偏差分类1 fenzi fenmu ratio	公里数偏差分类1 cnt ratio
0 2500-5000公里 56457 316906 0.178151	1 2500公里以内 412725 0.858953
1 2500公里以内 221241 316906 0.698128	0 2500-5000公里 35121 0.073093
2 5000-7500公里 15433 316906 0.048699	3 7500公里以上 20658 0.042993
3 7500公里以上 23775 316906 0.075022	2 5000-7500公里 11994 0.024962

Summaries

- 1. Business logical understanding
- 2. wide-format structure data design
- 3. Creative batch-wise training and statistics
- 4. Innovative multi-metric weighted fusion for predictions
- 5. Deploying with PySpark in production enables faster processing and improved scalability.

Innovations

- 1. Transformed the original long-format dataset into a wide-format structure to prevent data leakage during training, avoiding inflated model performance.
- 2. Fed 70% of well-cleaned data into the algorithmic model, while the remaining 30%—including cold-start users or data that could not be fully cleaned—were processed using a statistical rule-based model.
- 3. Both algorithmic and statistical rule-based models innovatively employed granular, batch-wise training and statistical analysis.
- 4. Within the Machine Learning model, an innovative multi-metric weighted fusion approach was applied, enabling strong performance even with a single model.

Overall Model Framework

